

5	\$1,048.53	\$1,023.91	\$1,000.00	\$976.77	\$954.20	\$932.27	\$910.96	\$890.25	\$870.12
10	\$1,094.71	\$1,046.11	\$1,000.00	\$956.24	\$914.70	\$875.25	\$837.78	\$802.18	\$768.35
20	\$1,180.46	\$1,085.84	\$1,000.00	\$922.05	\$851.23	\$786.81	\$728.19	\$674.80	\$626.13
30	\$1,258.08	\$1,120.08	\$1,000.00	\$895.35	\$804.00	\$724.12	\$654.16	\$592.78	\$538.83

Years to Maturity	New Interest Rate								
	1.0%	1.5%	2.0%	2.5%	3.0%	3.5%	4.0%	4.5%	5.0%
1	1.0%	0.5%	0.0%	−0.5%	−1.0%	−1.4%	−1.9%	−2.4%	−2.9%
5	4.9%	2.4%	0.0%	−2.3%	−4.6%	−6.8%	−8.9%	−11.0%	−13.0%
10	9.5%	4.6%	0.0%	−4.4%	−8.5%	−12.5%	−16.2%	−19.8%	−23.2%
20	18.0%	8.6%	0.0%	−7.8%	−14.9%	−21.3%	−27.2%	−32.5%	−37.4%
30	25.8%	12.0%	0.0%	−10.5%	−19.6%	−27.6%	−34.6%	−40.7%	−46.1%

straightforward way. A bond's duration is essentially the effective maturity of a bond—an average of when the bond's payments are received, weighted by the discounted size of those cash flows. A zero-coupon bond provides one payment at the maturity date, so its duration is the same as the time to maturity. The further away the maturity date, the higher the bond's duration, making it more sensitive to interest rate changes. A coupon bond will have a shorter duration than the time to the maturity date, because coupon payments are received before the maturity date. Higher coupon rates push relatively more cash flows sooner, which otherwise lowers the duration for a bond with the same maturity date. Also, lower interest rates mean the future cash flows from a bond are discounted less relative to nearer-term cash flows, and so bond duration increases when interest rates are low. An implication for this point is that our low interest- rate environment increases the interest rate risk for holding bonds, as a rate increase can result in a bigger capital loss.

An observant reader of Exhibit 7.2 might note that duration is not symmetric. For a thirty-year bond, a 1% increase in interest rates to 3% percent results in a capital loss of 19.6 percent, while a 1 percent decrease in interest rates to 1 percent results in a capital gain of 25.6 percent. The duration measure works best for small interest rate changes because it is a linear approximation to a